

WaypointGen Supplementary Material

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S.1 Independent Compositional Re-Scoring

Independent semantic LLM-judge re-score on all GT constraints: 422/713 = 59.2% overall (D1 43.8%, D2 55.7%, D3 73.8%, D4 91.1%). Different constraint enumeration than main Table 3 (713 vs 579); both methodologies reported for transparency.

S.2 Stage-2 LoRA Adapter

Base Qwen2.5-VL-7B = 0% area-equation validity (train + held-out, n=15+18); + Stage-2 LoRA = 86.7% train / 44.4% held-out / 63.6% overall. Validates the learned-component design. Caveats: validity = grammar well-formedness (not metric grounding); judge accepts LoRA-invented predicate variants; weakest on ≥ 3 simultaneous constraints.

S.3 Edge Latency on NVIDIA Jetson Thor

n=10 cmds spanning L0-L3 on Qwen3.6-35B-A3B: mean 9.4s/cmd (med 8.0), decode 35.2 tps, output 256 tokens avg, CV <0.05 (decode-bound on 122GB unified-memory edge). The full per-command breakdown is given in Table 1.

S.4 Integrity Reconciliation

Every headline metric was independently re-derived from the on-disk scored evaluation log (`v15_gt200_scored`, $n = 200$ pairs). Table 2 lists the authoritative values; the per-pair recompute in Table 3 reproduces them exactly. Where two enumeration methodologies exist (compositional satisfaction), both are reported (main paper vs. independent re-judge in S.1) for full transparency.

S.5 Per-Pair Evaluation Table

Table 3 reports every one of the 200 ground-truth NL $\rightarrow$ waypoint pairs (20 command categories $\times$ 4 detail levels D1–D4), grouped by detail level with a per-level subtotal. Structural pass measures whether a well-formed area-equation

Table 1: Per-command edge inference latency on the NVIDIA Jetson Thor (122 GB unified memory, Qwen3.6-35B-A3B), $n = 10$ commands spanning detail levels L0–L3. Wall = end-to-end wall-clock; tokens = output tokens; decode = decode throughput.

#	command	wall (s)	tokens	decode (tok/s)
0	Drive down row 3 corridor	22.15	256	35.2
1	Traverse all even rows from north end	7.98	256	35.3
2	Stop at the third tree from the gate	8.05	256	35.2
3	Avoid the missing tree at row 2 column 4	7.98	256	35.3
4	Go to the southwest quadrant	8.04	256	35.2
5	Enter the headland and circle clockwise	8.10	256	35.3
6	Pick up at tree (3, 5)	8.04	256	35.2
7	Skip every other row starting from row 1	8.04	256	35.2
8	Stay 2m from the trunk on the east side	8.03	256	35.2
9	Inspect the gap between rows 4 and 5	8.03	256	35.2
mean		9.44	256	35.2
median		8.04	–	35.2

and waypoint sequence were produced; geometric pass measures whether the generated waypoints fall inside the target region. The structural–geometric gap (99.0% vs. 39.5%) and its sharp dependence on target-region size (D1 80% $\rightarrow$ D4 6%) motivate the learned geometry encoder discussed in the main paper.

Table 3: Per-pair evaluation on the 200 ground-truth NL $\rightarrow$ waypoint pairs (20 command categories $\times$ 4 detail levels D1–D4), grouped by detail level. SR = success round (multi-round inference, 1–3); Struct = structural pass; n_{wp} = generated waypoints; f_{in} = fraction of waypoints inside the target region; Geo = geometric pass.

image_id	D	category	SR	Struct	n_{wp}	f_{in}	Geo
citrus_bev_sample_02	D1	centre_then_quadrant	1	✓	6	0.50	✓
citrus_bev_sample_03	D1	centre_then_quadrant	1	✓	6	0.71	✓
citrus_bev_sample_04	D1	centre_then_quadrant	1	✓	6	0.17	×
citrus_bev_sample_05	D1	centre_then_quadrant	1	✓	6	0.00	×
citrus_bev_sample_06	D1	centre_then_quadrant	3	✓	6	0.50	✓
citrus_bev_sample_07	D1	centre_then_quadrant	1	✓	6	1.00	✓
citrus_bev_sample_08	D1	centre_then_quadrant	1	✓	6	0.00	×
citrus_bev_sample_09	D1	centre_then_quadrant	2	✓	6	0.33	×
citrus_bev_sample_10	D1	centre_then_quadrant	1	✓	6	0.17	×
citrus_bev_sample_11	D1	centre_then_quadrant	2	✓	6	0.33	×
citrus_bev_sample_02	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_03	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_04	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_05	D1	multi_constraint_avoidance	2	✓	6	1.00	✓
citrus_bev_sample_06	D1	multi_constraint_avoidance	2	✓	6	1.00	✓
citrus_bev_sample_07	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_08	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_09	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_10	D1	multi_constraint_avoidance	1	✓	6	0.67	✓

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Table 3 – continued

image_id	D	category	SR	Struct	n_{wp}	f_{in}	Geo
citrus_bev_sample_11	D1	multi_constraint_avoidance	1	✓	6	1.00	✓
citrus_bev_sample_02	D1	northern_half_then_southern_half	1	✓	6	0.78	✓
citrus_bev_sample_03	D1	northern_half_then_southern_half	2	✓	6	1.00	✓
citrus_bev_sample_04	D1	northern_half_then_southern_half	1	✓	6	0.67	✓
citrus_bev_sample_05	D1	northern_half_then_southern_half	1	✓	6	1.00	✓
citrus_bev_sample_06	D1	northern_half_then_southern_half	1	✓	6	0.69	✓
citrus_bev_sample_07	D1	northern_half_then_southern_half	1	✓	6	1.00	✓
citrus_bev_sample_08	D1	northern_half_then_southern_half	2	✓	6	1.00	✓
citrus_bev_sample_09	D1	northern_half_then_southern_half	2	✓	6	1.00	✓
citrus_bev_sample_10	D1	northern_half_then_southern_half	1	✓	6	1.00	✓
citrus_bev_sample_11	D1	northern_half_then_southern_half	3	×	6	0.00	×
citrus_bev_sample_02	D1	perimeter_loop	1	✓	6	1.00	✓
citrus_bev_sample_03	D1	perimeter_loop	1	✓	6	1.00	✓
citrus_bev_sample_04	D1	perimeter_loop	2	✓	6	1.00	✓
citrus_bev_sample_05	D1	perimeter_loop	1	✓	6	1.00	✓
citrus_bev_sample_06	D1	perimeter_loop	2	✓	6	1.00	✓
citrus_bev_sample_07	D1	perimeter_loop	2	✓	6	1.00	✓
citrus_bev_sample_08	D1	perimeter_loop	2	✓	6	1.00	✓
citrus_bev_sample_09	D1	perimeter_loop	1	✓	6	1.00	✓
citrus_bev_sample_10	D1	perimeter_loop	1	✓	6	1.00	✓
citrus_bev_sample_11	D1	perimeter_loop	2	✓	6	1.00	✓
citrus_bev_sample_02	D1	quadrant_coverage	1	✓	6	1.00	✓
citrus_bev_sample_03	D1	quadrant_coverage	1	✓	6	1.00	✓
citrus_bev_sample_04	D1	quadrant_coverage	2	✓	6	0.33	×
citrus_bev_sample_05	D1	quadrant_coverage	1	✓	6	0.20	×
citrus_bev_sample_06	D1	quadrant_coverage	2	✓	6	1.00	✓
citrus_bev_sample_07	D1	quadrant_coverage	1	✓	6	1.00	✓
citrus_bev_sample_08	D1	quadrant_coverage	1	×	6	0.50	✓
citrus_bev_sample_09	D1	quadrant_coverage	2	✓	6	1.00	✓
citrus_bev_sample_10	D1	quadrant_coverage	1	✓	6	0.33	×
citrus_bev_sample_11	D1	quadrant_coverage	2	✓	6	0.50	✓
D1 subtotal ($n = 50$)			48/50		40/50		
citrus_bev_sample_02	D2	alternating_rows_zig_zag	2	✓	6	0.33	×
citrus_bev_sample_03	D2	alternating_rows_zig_zag	1	✓	6	0.33	×
citrus_bev_sample_04	D2	alternating_rows_zig_zag	1	✓	6	0.00	×
citrus_bev_sample_05	D2	alternating_rows_zig_zag	1	✓	6	0.00	×
citrus_bev_sample_06	D2	alternating_rows_zig_zag	1	✓	6	1.00	✓
citrus_bev_sample_07	D2	alternating_rows_zig_zag	1	✓	6	0.33	×
citrus_bev_sample_08	D2	alternating_rows_zig_zag	1	✓	4	0.00	×
citrus_bev_sample_09	D2	alternating_rows_zig_zag	1	✓	6	0.00	×
citrus_bev_sample_10	D2	alternating_rows_zig_zag	1	✓	6	0.67	✓
citrus_bev_sample_11	D2	alternating_rows_zig_zag	1	✓	6	0.00	×
citrus_bev_sample_02	D2	every_other_tree_multi_row	1	✓	6	0.67	✓
citrus_bev_sample_03	D2	every_other_tree_multi_row	2	✓	6	0.67	✓
citrus_bev_sample_04	D2	every_other_tree_multi_row	2	✓	6	0.67	✓
citrus_bev_sample_05	D2	every_other_tree_multi_row	2	✓	6	1.00	✓
citrus_bev_sample_06	D2	every_other_tree_multi_row	1	✓	6	0.00	×
citrus_bev_sample_07	D2	every_other_tree_multi_row	2	✓	6	0.33	×
citrus_bev_sample_08	D2	every_other_tree_multi_row	1	✓	6	0.00	×
citrus_bev_sample_09	D2	every_other_tree_multi_row	1	✓	6	0.20	×
citrus_bev_sample_10	D2	every_other_tree_multi_row	3	✓	6	1.00	✓
citrus_bev_sample_11	D2	every_other_tree_multi_row	1	✓	6	0.00	×
citrus_bev_sample_02	D2	headland_strip_sweep	1	✓	6	0.00	×

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Table 3 – continued

image_id	D	category	SR	Struct	n_{wp}	f_{in}	Geo
citrus_bev_sample_03	D2	headland_strip_sweep	1	✓	6	0.00	×
citrus_bev_sample_04	D2	headland_strip_sweep	3	✓	6	0.00	×
citrus_bev_sample_05	D2	headland_strip_sweep	1	✓	6	0.00	×
citrus_bev_sample_06	D2	headland_strip_sweep	2	✓	6	0.00	×
citrus_bev_sample_07	D2	headland_strip_sweep	1	✓	6	0.00	×
citrus_bev_sample_08	D2	headland_strip_sweep	1	✓	6	0.00	×
citrus_bev_sample_09	D2	headland_strip_sweep	1	✓	6	0.00	×
citrus_bev_sample_10	D2	headland_strip_sweep	1	✓	6	0.00	×
citrus_bev_sample_11	D2	headland_strip_sweep	2	✓	6	0.00	×
citrus_bev_sample_02	D2	odd_rows_only	1	✓	6	1.00	✓
citrus_bev_sample_03	D2	odd_rows_only	1	✓	6	0.50	✓
citrus_bev_sample_04	D2	odd_rows_only	1	✓	6	0.75	✓
citrus_bev_sample_05	D2	odd_rows_only	1	✓	6	0.44	×
citrus_bev_sample_06	D2	odd_rows_only	1	✓	6	0.75	✓
citrus_bev_sample_07	D2	odd_rows_only	1	✓	6	0.75	✓
citrus_bev_sample_08	D2	odd_rows_only	2	✓	6	0.33	×
citrus_bev_sample_09	D2	odd_rows_only	1	✓	6	1.00	✓
citrus_bev_sample_10	D2	odd_rows_only	2	✓	6	1.00	✓
citrus_bev_sample_11	D2	odd_rows_only	2	✓	6	1.00	✓
citrus_bev_sample_02	D2	row_pair_with_corridor_transition	1	✓	6	1.00	✓
citrus_bev_sample_03	D2	row_pair_with_corridor_transition	1	✓	6	1.00	✓
citrus_bev_sample_04	D2	row_pair_with_corridor_transition	1	✓	4	1.00	✓
citrus_bev_sample_05	D2	row_pair_with_corridor_transition	1	✓	6	1.00	✓
citrus_bev_sample_06	D2	row_pair_with_corridor_transition	1	✓	6	0.00	×
citrus_bev_sample_07	D2	row_pair_with_corridor_transition	1	✓	6	0.67	✓
citrus_bev_sample_08	D2	row_pair_with_corridor_transition	1	✓	6	0.00	×
citrus_bev_sample_09	D2	row_pair_with_corridor_transition	1	✓	6	0.50	✓
citrus_bev_sample_10	D2	row_pair_with_corridor_transition	1	✓	6	1.00	✓
citrus_bev_sample_11	D2	row_pair_with_corridor_transition	1	✓	6	0.00	×
D2 subtotal ($n = 50$)			50/50		22/50		
citrus_bev_sample_02	D3	headland_then_row_half	1	✓	6	0.33	×
citrus_bev_sample_03	D3	headland_then_row_half	2	✓	6	0.50	✓
citrus_bev_sample_04	D3	headland_then_row_half	1	✓	6	0.33	×
citrus_bev_sample_05	D3	headland_then_row_half	1	✓	6	0.33	×
citrus_bev_sample_06	D3	headland_then_row_half	1	✓	6	0.00	×
citrus_bev_sample_07	D3	headland_then_row_half	2	✓	6	0.33	×
citrus_bev_sample_08	D3	headland_then_row_half	2	✓	6	0.33	×
citrus_bev_sample_09	D3	headland_then_row_half	2	✓	6	0.33	×
citrus_bev_sample_10	D3	headland_then_row_half	2	✓	6	0.50	✓
citrus_bev_sample_11	D3	headland_then_row_half	1	✓	6	0.17	×
citrus_bev_sample_02	D3	lane_then_row	1	✓	6	0.17	×
citrus_bev_sample_03	D3	lane_then_row	2	✓	6	1.00	✓
citrus_bev_sample_04	D3	lane_then_row	1	✓	6	0.17	×
citrus_bev_sample_05	D3	lane_then_row	1	✓	6	0.00	×
citrus_bev_sample_06	D3	lane_then_row	1	✓	6	0.17	×
citrus_bev_sample_07	D3	lane_then_row	1	✓	6	0.17	×
citrus_bev_sample_08	D3	lane_then_row	1	✓	6	0.17	×
citrus_bev_sample_09	D3	lane_then_row	1	✓	6	0.17	×
citrus_bev_sample_10	D3	lane_then_row	1	✓	6	0.33	×
citrus_bev_sample_11	D3	lane_then_row	1	✓	6	0.38	×
citrus_bev_sample_02	D3	row_central_segment	1	✓	6	0.83	✓
citrus_bev_sample_03	D3	row_central_segment	1	✓	6	0.00	×
citrus_bev_sample_04	D3	row_central_segment	1	✓	6	1.00	✓

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Table 3 – continued

image_id	D	category	SR	Struct	n_{wp}	f_{in}	Geo
citrus_bev_sample_05	D3	row_central_segment	1	✓	6	1.00	✓
citrus_bev_sample_06	D3	row_central_segment	2	✓	6	0.00	×
citrus_bev_sample_07	D3	row_central_segment	1	✓	5	1.00	✓
citrus_bev_sample_08	D3	row_central_segment	1	✓	6	1.00	✓
citrus_bev_sample_09	D3	row_central_segment	1	✓	6	0.00	×
citrus_bev_sample_10	D3	row_central_segment	1	✓	6	0.50	✓
citrus_bev_sample_11	D3	row_central_segment	1	✓	6	1.00	✓
citrus_bev_sample_02	D3	row_then_headland	1	✓	6	0.11	×
citrus_bev_sample_03	D3	row_then_headland	1	✓	6	0.83	✓
citrus_bev_sample_04	D3	row_then_headland	1	✓	6	0.00	×
citrus_bev_sample_05	D3	row_then_headland	1	✓	6	0.00	×
citrus_bev_sample_06	D3	row_then_headland	1	✓	6	0.83	✓
citrus_bev_sample_07	D3	row_then_headland	1	✓	6	0.33	×
citrus_bev_sample_08	D3	row_then_headland	1	✓	6	0.17	×
citrus_bev_sample_09	D3	row_then_headland	1	✓	6	0.17	×
citrus_bev_sample_10	D3	row_then_headland	1	✓	6	0.33	×
citrus_bev_sample_11	D3	row_then_headland	1	✓	6	0.50	✓
citrus_bev_sample_02	D3	row_with_inline_avoid	1	✓	6	0.17	×
citrus_bev_sample_03	D3	row_with_inline_avoid	1	✓	6	0.33	×
citrus_bev_sample_04	D3	row_with_inline_avoid	1	✓	6	0.00	×
citrus_bev_sample_05	D3	row_with_inline_avoid	1	✓	6	0.67	✓
citrus_bev_sample_06	D3	row_with_inline_avoid	2	✓	6	0.00	×
citrus_bev_sample_07	D3	row_with_inline_avoid	1	✓	6	0.00	×
citrus_bev_sample_08	D3	row_with_inline_avoid	1	✓	6	0.33	×
citrus_bev_sample_09	D3	row_with_inline_avoid	1	✓	6	0.33	×
citrus_bev_sample_10	D3	row_with_inline_avoid	2	✓	6	0.00	×
citrus_bev_sample_11	D3	row_with_inline_avoid	1	✓	6	0.33	×
D3 subtotal ($n = 50$)			50/50		14/50		
citrus_bev_sample_02	D4	single_tree_inspection	2	✓	6	0.00	×
citrus_bev_sample_03	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_04	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_05	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_06	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_07	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_08	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_09	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_10	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_11	D4	single_tree_inspection	1	✓	6	0.00	×
citrus_bev_sample_02	D4	single_tree_landmark_skip	2	✓	6	0.00	×
citrus_bev_sample_03	D4	single_tree_landmark_skip	1	✓	6	0.00	×
citrus_bev_sample_04	D4	single_tree_landmark_skip	1	✓	6	0.17	×
citrus_bev_sample_05	D4	single_tree_landmark_skip	1	✓	6	0.00	×
citrus_bev_sample_06	D4	single_tree_landmark_skip	1	✓	6	0.00	×
citrus_bev_sample_07	D4	single_tree_landmark_skip	2	✓	6	0.00	×
citrus_bev_sample_08	D4	single_tree_landmark_skip	1	✓	6	0.33	×
citrus_bev_sample_09	D4	single_tree_landmark_skip	1	✓	6	0.17	×
citrus_bev_sample_10	D4	single_tree_landmark_skip	1	✓	6	0.00	×
citrus_bev_sample_11	D4	single_tree_landmark_skip	1	✓	6	0.00	×
citrus_bev_sample_02	D4	single_tree_pickup	2	✓	6	0.00	×
citrus_bev_sample_03	D4	single_tree_pickup	1	✓	6	0.00	×
citrus_bev_sample_04	D4	single_tree_pickup	1	✓	6	0.00	×
citrus_bev_sample_05	D4	single_tree_pickup	1	✓	6	0.00	×
citrus_bev_sample_06	D4	single_tree_pickup	1	✓	6	0.00	×

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Table 3 – continued

image_id	D	category	SR	Struct	n_{wp}	f_{in}	Geo
citrus_bev_sample_07	D4	single_tree_pickup	1	✓	6	0.00	×
citrus_bev_sample_08	D4	single_tree_pickup	2	✓	6	0.00	×
citrus_bev_sample_09	D4	single_tree_pickup	1	✓	6	0.00	×
citrus_bev_sample_10	D4	single_tree_pickup	1	✓	6	0.00	×
citrus_bev_sample_11	D4	single_tree_pickup	2	✓	6	0.00	×
citrus_bev_sample_02	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_03	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_04	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_05	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_06	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_07	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_08	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_09	D4	single_tree_replanting	2	✓	6	0.00	×
citrus_bev_sample_10	D4	single_tree_replanting	1	✓	6	0.17	×
citrus_bev_sample_11	D4	single_tree_replanting	1	✓	6	0.00	×
citrus_bev_sample_02	D4	single_tree_via_corridor	1	✓	6	0.50	✓
citrus_bev_sample_03	D4	single_tree_via_corridor	1	✓	6	0.17	×
citrus_bev_sample_04	D4	single_tree_via_corridor	1	✓	6	0.00	×
citrus_bev_sample_05	D4	single_tree_via_corridor	1	✓	6	0.50	✓
citrus_bev_sample_06	D4	single_tree_via_corridor	3	✓	6	0.00	×
citrus_bev_sample_07	D4	single_tree_via_corridor	1	✓	6	0.50	✓
citrus_bev_sample_08	D4	single_tree_via_corridor	1	✓	6	0.33	×
citrus_bev_sample_09	D4	single_tree_via_corridor	1	✓	6	0.17	×
citrus_bev_sample_10	D4	single_tree_via_corridor	1	✓	6	0.09	×
citrus_bev_sample_11	D4	single_tree_via_corridor	1	✓	6	0.17	×
D4 subtotal ($n = 50$)			50/50			3/50	
Overall ($n = 200$)			198/200			79/200	

S.6 Reproducibility

Code + 200-pair eval set + 6-shot ICL bank + raw inference logs released on acceptance. Hardware: RTX 5090, NVIDIA GB10, NVIDIA Jetson Thor.

Table 2: Integrity reconciliation: every headline metric re-derived from the on-disk scored evaluation log (`v15_gt200_scored`, $n = 200$). Per-pair recompute (Table 3) agrees with these authoritative values.

metric	authoritative value
Structural output rate	200/200 = 100.0%
Structural pass rate	198/200 = 99.0%
≥ 6 waypoints	197/200 = 98.5%
Geometric pass (overall)	79/200 = 39.5%
Geometric pass D1 / D2 / D3 / D4	80% / 44% / 28% / 6%
Structural pass D1 / D2 / D3 / D4	96% / 100% / 100% / 100%
Round-1 success (single-round baseline)	153/200 = 76.5%
Success-round distribution {1,2,3}	153 / 42 / 5
RTX 5090 latency (median / mean)	28.0 / 28.8 s/cmd
Jetson Thor edge structural	199/200 = 99.5%
Compositional satisfaction (independent re-judge)	422/713 = 59.2%